

α : one value per item

	noun	verb
walk	α_1	α_1
run	α_2	α_2
$\vdots$	$\vdots$	$\vdots$
word l	α_l	α_l

each repeated C times

β : one value per level

	noun	verb
walk	β_1	β_2
run	β_1	β_2
$\vdots$	$\vdots$	$\vdots$
word l	β_1	β_2

each repeated l times

γ : one value per position

	noun	verb
walk	γ_{11}	γ_{12}
run	γ_{21}	γ_{22}
$\vdots$	$\vdots$	$\vdots$
word l	γ_{l1}	γ_{l2}

never repeated

$$\sum_{\{i,c\}} \|M[i,c] - \mu\|^2 = C \cdot \sum_i \|\alpha_i\|^2 + l \cdot \sum_c \|\beta_c\|^2 + \sum_{\{i,c\}} \|\gamma_{ic}\|^2$$